Photosynthesis and ecosystems

Green plants produce their own food using photosynthesis. This food helps them grow. As the plant grows, more leaves, fruit and flowers are made available for other living things to eat. In this way, photosynthesis helps an ecosystem to thrive.

Plants use carbon dioxide, water and the sun's energy to make a sugary food called glucose. When the plant is eaten by a consumer, some of the energy stored in the glucose is released and used by the consumer. The rest of the energy is released as heat. When consumers and producers die, decomposers break down their remains into tiny parts called nutrients. These nutrients go into the soil or the atmosphere. A new plant uses these nutrients to begin the **energy transfer process** all over again.

☞ **1** Write the words in the word bank in alphabetical order.

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_____

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_____

Word bank
photosynthesis
ecosystem
carbon dioxide
water
energy
glucose
nutrients

2 **a)** There are two parts to the word *photosynthesis*. *Photo* means 'light' and *synthesis* means 'to make'. What do plants use light to make?

_____

b) Other English words use the prefix (word beginning) *photo*. List at least three.

_____ _____

_____ _____

3 In your own words, describe the function of photosynthesis in ecosystems.

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_____

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Photosynthesis and ecosystems CONT'D

3 Photosynthesis occurs in green plants. Cross out the letters p h o t o s y n t h e s i s, in order, to find the names of the plant parts that do each job. Write the plant parts in the table. The first one has been done for you.

Jobs	**Plant part**
collect carbon dioxide	phloetaovsyntheesiss leaves
attract insects for pollination (the process that allows some plants to reproduce)	fphlotosowyntheseriss _____
store seeds	pfhotorusyinthesist _____
provide support and strength	phsotostynthaesislk _____
collect water	phrootoosynthesists _____

4 Label the diagram below with the names of the plant parts from Question 3.



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